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I want to create a python codebase where I have an app which will look through banking statements (for debit cards and credit cards), and it will create a .csv file which creates two tables. One table is for income, and the other is for expenses. These tables should have their own categories for each type of transaction recorded. What kind of libraries should I look at to achieve this?

That's actually a really solid project idea, Ethan — especially with your Python + data background. It's practical, resume-worthy, and expandable (analytics, dashboards, ML categorization later).

Here's how I'd break down the **library stack**, depending on what stage you're at.

1️⃣ Reading Bank Statements

Bank statements usually come as:

- **PDF**
- **CSV**
- **Excel (.xlsx)**
- Sometimes scraped from web exports

📄 If Statements Are PDFs

Look into:

- `pdfplumber` → great for extracting structured text/tables from PDFs
- `PyPDF2` → basic PDF text extraction
- `tabula-py` → extracts tables from PDFs (works best if tables are clean)

👉 If they're scanned PDFs (images), you'll need:

- `pytesseract` (OCR)
- `Pillow` (image preprocessing)

📊 If Statements Are CSV or Excel

Much easier.

- `pandas` → **this will be your core library**
- `openpyxl` → for Excel (.xlsx)

Example:

```python
import pandas as pd

df = pd.read_csv("statement.csv")
```

2️⃣ Data Cleaning & Transformation

This is where most of the work happens.

You'll want:

- `pandas` → filtering, grouping, aggregating
- `re` (built-in regex) → pattern matching transaction descriptions
- `datetime` → date normalization

You'll likely:

- Standardize columns (Date, Description, Amount)
- Convert debit/credit into positive/negative
- Normalize merchant names
- Remove duplicates

3️⃣ Categorizing Transactions

This is the interesting part.

Option A — Rule-Based (Recommended First)

Use:

- `re`

- Python dictionaries

Example:

Python

```python
CATEGORY_MAP = {
    "groceries": ["WALMART", "LOBLAWS", "COSTCO"],
    "gas": ["SHELL", "ESSO"],
    "restaurants": ["UBER EATS", "DOORDASH"]
}
```

Then loop and assign category based on keyword matching.

This is:

- Simple
- Fast
- Easy to debug
- Resume-friendly

Option B — Smarter Categorization (Later Upgrade)

If you want something more advanced:

- `scikit-learn` → classification model
- `nltk` or `spaCy` → NLP
- `fuzzywuzzy` or `rapidfuzz` → fuzzy string matching

This lets you:

- Auto-learn categories
- Handle messy merchant names
- Reduce manual rules

But start rule-based.

4️⃣ Splitting Income vs Expenses

With `pandas`, this is easy:

Python

```python
income_df = df[df["Amount"] > 0]
expense_df = df[df["Amount"] < 0]
```

Then:

Python

```python
income_df.to_csv("income.csv", index=False)
expense_df.to_csv("expenses.csv", index=False)
```

If you want **both tables in one CSV**, that's not standard CSV behavior (CSV = single table).

Instead, you should:

- Use Excel with multiple sheets (`pandas.ExcelWriter`)
- OR create two separate CSV fi